DAY-3 Computer Networks Notes

Topics: NAT, Subnetting, Session Layer, Transport Layer, Presentation Layer

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1. NAT (Network Address Translation)

Definition

NAT stands for **Network Address Translation**.
It is a technique used in networking in which a router or gateway changes the **private IP address** of a device into a **public IP address** before sending data to the internet.

In simple words, NAT helps many devices inside a private network use **one public IP address** to communicate with the outside world.

Why NAT is needed

IPv4 addresses are limited.
If every device in the world needed a unique public IP, IPv4 addresses would finish quickly.

So, inside homes, offices, colleges, etc., devices are given **private IP addresses** such as:

- 192.168.x.x
- 10.x.x.x
- 172.16.x.x to 172.31.x.x

These private IP addresses cannot directly work on the internet.
Hence, NAT is used to convert them into a public IP.

Example of NAT

Suppose in your home:

- Laptop IP = 192.168.1.2
- Phone IP = 192.168.1.3
- Smart TV IP = 192.168.1.4

Your router has:

- Public IP = 49.36.120.10

When your laptop sends a request to Google:

- Source private IP = 192.168.1.2
- Router changes it to public IP = 49.36.120.10

So Google sees the request as coming from **49.36.120.10**, not from 192.168.1.2.

When the response comes back, the router checks its NAT table and forwards the response to the correct device.

How NAT works

1. A device sends data to the router.
2. The router replaces the private IP with its public IP.
3. The router stores the mapping in a NAT table.
4. The packet goes to the internet.
5. Response comes back to the router.
6. Router uses the NAT table to send data to the correct internal device.

NAT Table

A NAT device maintains a table like this:

Private IP	Private Port	Public IP	Public Port
192.168.1.2	5001	49.36.120.10	3001
192.168.1.3	5002	49.36.120.10	3002

This allows many devices to use the same public IP.

Types of NAT

1. Static NAT

In static NAT, one private IP is permanently mapped to one public IP.

Example

- 192.168.1.10 → 122.10.5.7

Use

Used when a device inside the network must always be reachable from outside, such as a server.

Advantage

- Fixed mapping
- Easy for servers

Disadvantage

- Requires more public IPs

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2. Dynamic NAT

In dynamic NAT, a private IP is mapped to any available public IP from a pool of public addresses.

Example

- 192.168.1.2 may get 122.10.5.7
- 192.168.1.3 may get 122.10.5.8

Advantage

- Flexible use of public IP pool

Disadvantage

- Still needs multiple public IPs

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3. PAT (Port Address Translation) / NAT Overload

This is the most common type of NAT.
Many private IP addresses share one public IP address using different port numbers.

Example

- Laptop → 49.36.120.10:3001
- Phone → 49.36.120.10:3002
- TV → 49.36.120.10:3003

Advantage

- Saves public IP addresses
- Used in homes and offices

Disadvantage

- Slightly more complexity

Advantages of NAT

1. Saves public IPv4 addresses
2. Allows multiple devices to share one public IP
3. Adds a basic level of security by hiding private IPs
4. Makes internal addressing easy to manage

Disadvantages of NAT

1. Adds processing at the router
2. Can cause issues in some real-time applications
3. End-to-end communication becomes harder
4. Some protocols need special handling

NAT and Security

NAT is **not a firewall**, but it gives some protection because private addresses are hidden from the outside world.
However, full security still needs:

- Firewall
- IDS/IPS
- Access control

One-line answer

NAT is a technique in which a router converts private IP addresses into public IP addresses so devices in a private network can access the internet.

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2. Subnetting

Definition

Subnetting is the process of dividing one large network into multiple smaller networks called **subnets**.

It improves:

- network management
- performance
- security
- efficient IP usage

Why subnetting is needed

If one network is very large:

- there will be too much broadcast traffic
- management becomes difficult
- IP addresses may get wasted

Subnetting solves this by creating smaller networks.

Important terms in subnetting

1. IP Address

An IP address identifies a device in a network.

Example IPv4:

- 192.168.1.10

IPv4 has **32 bits**.

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2. Network ID

It identifies the network portion of the IP address.

Example:
In 192.168.1.10/24

- Network ID = 192.168.1.0

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3. Host ID

It identifies the specific device in that network.

In 192.168.1.10/24

- Host part = .10

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4. Subnet Mask

A subnet mask tells which part of the IP is network and which part is host.

Example:

- 255.255.255.0 = /24

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5. CIDR notation

CIDR writes subnet mask in short form.

Examples:

- /8
- /16
- /24
- /26

The number tells how many bits are used for the network.

Example of subnetting

Suppose we have a network:

- 192.168.1.0/24

A /24 network has:

- 24 bits for network
- 8 bits for host

Total addresses:

- 2^8 = 256

Usable host addresses:

- 256 - 2 = 254

Why minus 2?

- 1 for network address
- 1 for broadcast address

So usable hosts = **254**

Need to divide into 4 subnets

If we divide 192.168.1.0/24 into 4 subnets:
We borrow 2 bits from host part.

So new subnet:

- /26

Because:

- /24 + 2 = /26

Each subnet has:

- 2^(32-26) = 64 addresses
- usable hosts = 62

4 subnets are:

1. 192.168.1.0/26
   Host range: 192.168.1.1 to 192.168.1.62
   Broadcast: 192.168.1.63
2. 192.168.1.64/26
   Host range: 192.168.1.65 to 192.168.1.126
   Broadcast: 192.168.1.127
3. 192.168.1.128/26
   Host range: 192.168.1.129 to 192.168.1.190
   Broadcast: 192.168.1.191
4. 192.168.1.192/26
   Host range: 192.168.1.193 to 192.168.1.254
   Broadcast: 192.168.1.255

Formula used in subnetting

Number of subnets

If we borrow **n bits**, then:
Number of subnets = 2^n

Number of hosts per subnet

If **h bits** remain for host:
Hosts per subnet = 2^h - 2

Benefits of subnetting

1. Reduces broadcast traffic
2. Improves performance
3. Better security and isolation
4. Efficient use of IP addresses
5. Easier management

Drawback of subnetting

- Requires planning
- Misconfiguration can cause network problems

Easy memory example

Think of a company building.

- Whole building = one big network
- Each floor = one subnet
- Each room = one host

This makes management easier.

One-line answer

Subnetting is the process of dividing a large network into smaller sub-networks to improve performance, security, and efficient IP address usage.

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3. Session Layer

Definition

The **Session Layer** is the **5th layer of the OSI model**.
It is responsible for **establishing, managing, and terminating sessions** between two communicating devices or applications.

Simple meaning

It manages the conversation between two systems.

Just like in a phone call:

- call starts
- call continues
- call ends

That is what the session layer does for communication.

Functions of Session Layer

1. Session Establishment

It creates a session between sender and receiver before communication begins.

Example:
When two applications start communicating, a session is opened.

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2. Session Management

It keeps the session active and organized during communication.

It controls:

- whose turn to send
- whether the session is active
- whether the session is interrupted

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3. Session Termination

It closes the session properly when communication ends.

Example:
When you log out from a service or end a remote session.

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4. Synchronization

The session layer adds **checkpoints** during long transmissions.

If failure happens, communication resumes from the last checkpoint instead of restarting from zero.

Example

If a large file transfer stops at 90%, it can continue from the checkpoint.

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5. Dialog Control

It decides the mode of communication:

- Half duplex: one side talks at a time
- Full duplex: both sides talk together

Importance of Session Layer

Without session management:

- communication can become unorganized
- interruptions can cause restart from beginning
- long data exchange becomes harder

Example in real life

Suppose you are in a video call:

- Session starts when call begins
- Maintained while both people talk
- Ends when call disconnects

This is a simple real-life analogy of session layer.

Session layer services

- login sessions
- remote procedure calls
- dialog control
- synchronization in large transfers

One-line answer

Session layer establishes, manages, and terminates sessions between applications and provides synchronization during communication.

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4. Transport Layer

Definition

The **Transport Layer** is the **4th layer of the OSI model**.
It provides **end-to-end communication** between applications on different devices.

Simple meaning

Network layer sends data from one device to another device.
Transport layer sends data from one application to another application.

For example:

- IP address finds the destination computer
- Port number finds the destination application

Main functions of Transport Layer

1. Process-to-process delivery

It delivers data to the correct process/application using **port numbers**.

Examples:

- browser
- email app
- WhatsApp
- FTP client

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2. Segmentation and Reassembly

Large data is divided into smaller pieces called **segments** before sending.
At the receiving side, the segments are joined back correctly.

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3. Port Addressing

Every application is identified by a port number.

Examples:

- HTTP → 80
- HTTPS → 443
- FTP → 21
- SMTP → 25
- DNS → 53

Port numbers help transport layer deliver data to the correct application.

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4. Connection Control

Transport layer provides:

- **Connection-oriented service**
- **Connectionless service**

Connection-oriented

Connection is established first, then data is sent.
Reliable and ordered.

Used by **TCP**

Connectionless

No connection setup. Data is sent directly.
Fast but unreliable.

Used by **UDP**

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5. Flow Control

Flow control prevents a fast sender from overwhelming a slow receiver.

It controls how much data can be sent at one time.

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6. Error Control

Transport layer checks for:

- lost data
- duplicate data
- damaged data
- out-of-order delivery

TCP provides error checking and retransmission.

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7. Multiplexing and Demultiplexing

Multiplexing

Data from many applications is collected and sent.

Demultiplexing

Received data is separated and sent to the correct application.

Transport Layer Protocols

TCP (Transmission Control Protocol)

Features

- connection-oriented
- reliable
- ordered delivery
- flow control
- error control
- retransmission
- congestion control

Use cases

- web browsing

- file transfer
- email
- banking

UDP (User Datagram Protocol)

Features

- connectionless
- fast
- less overhead
- no guarantee of delivery
- no guarantee of order

Use cases

- live streaming
- online gaming
- DNS
- voice/video calls

TCP vs UDP

TCP	UDP
Reliable	Unreliable
Connection-oriented	Connectionless
Ordered delivery	No order guarantee
Slower	Faster
More overhead	Less overhead

Important terms in transport layer

Segment

Transport layer PDU in TCP is called a **segment**.

Datagram

Transport layer PDU in UDP is called a **datagram**.

Socket

Combination of:

- IP address
- Port number

Example:

- 192.168.1.10:443

TCP 3-way handshake

TCP connection is established using:

1. SYN
2. SYN-ACK
3. ACK

This is called the **three-way handshake**.

One-line answer

Transport layer provides end-to-end process-to-process communication, segmentation, flow control, error control, and uses TCP and UDP protocols.

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5. Presentation Layer

Definition

The **Presentation Layer** is the **6th layer of the OSI model**.
It is responsible for **data translation, encryption, and compression**.

Simple meaning

It makes sure that data sent by one system can be understood by the other system.

It acts like a **translator** between application layer and the rest of the network.

Main functions of Presentation Layer

1. Translation

Different systems may use different data formats.
Presentation layer converts data into a common format.

Example

- ASCII to Unicode
- EBCDIC to ASCII

It helps devices with different representations understand the same data.

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2. Encryption

It converts plain data into unreadable form for security.

At receiver side, it decrypts the data back into readable form.

Example

Secure communication over HTTPS uses encryption concepts.

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3. Compression

It reduces the size of data before transmission to save bandwidth and time.

At receiver side, data is decompressed.

Example

Compressed images, videos, files

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4. Data Formatting

It handles the format and representation of data:

- text
- image
- audio
- video

Importance of Presentation Layer

Without it:

- systems with different formats may not understand each other
- secure encrypted communication becomes difficult
- transmission may take more bandwidth

Examples of functions

- character code conversion
- image format handling
- encrypted data handling
- compressed file handling

Real-life analogy

Suppose two people speak different languages.
A translator helps them understand each other.

That translator is like the **presentation layer**.

Examples related to presentation layer

- JPEG, PNG for images
- MP3 for audio
- MPEG for video
- SSL/TLS concepts linked with encryption support

One-line answer

Presentation layer translates, encrypts, and compresses data so that different systems can understand and securely exchange information.

Difference between Session, Transport and Presentation Layer

Layer	OSI Number	Main Work
Transport Layer	4	End-to-end delivery, segmentation, port addressing, reliability
Session Layer	5	Establishes, manages, and terminates sessions
Presentation Layer	6	Translation, encryption, compression

Very Short Revision Points

NAT

- Converts private IP to public IP
- Done by router
- Saves IPv4 addresses
- Types: Static, Dynamic, PAT

Subnetting

- Divides large network into smaller subnets
- Uses subnet mask
- Improves management and performance

Session Layer

- Starts, manages, ends sessions
- Provides synchronization
- OSI Layer 5

Transport Layer

- End-to-end communication
- Uses ports
- Protocols: TCP, UDP
- OSI Layer 4

Presentation Layer

- Translation
- Encryption
- Compression
- OSI Layer 6